SOP-101 — Centrifugal Pump Startup and Shutdown Procedure

Plant Site A · Rev 4 · Issued 12/03/2025 · Approved by Plant Engineering

1. Purpose

This procedure defines the approved method for starting up and shutting down the centrifugal process pumps installed in the Process Area and the Utilities Area of Plant Site A. Correct startup practice protects the mechanical seal, the bearings and the driver from avoidable damage, and ensures that flow is established in a controlled manner without water hammer or cavitation. Shutdown practice ensures the pump is left in a safe, drained or preserved state.

2. Scope

This procedure applies to all centrifugal pumps in the plant equipment register, including Pump 101A, Pump 101B and Pump 101C in the Process Area, and the cooling water pumps P-102A, P-102B and P-103A in the Utilities Area. It covers routine operational starts and planned shutdowns. It does not cover first commissioning, post-overhaul acceptance testing, or emergency trips, which are addressed in the relevant maintenance and operations manuals.

3. References

SOP-114 Centrifugal Pump Suction Strainer Maintenance.

SOP-110 Rotating Equipment Vibration Monitoring and Bearing Lubrication.

SOP-118 Equipment Isolation and Lock-Out Tag-Out.

Manufacturer operating manual for the installed pump model, held in the technical library.

4. Safety Precautions

Confirm that no isolation tags or locks are present on the pump, its driver or its switchgear before attempting a start. If a tag is present, stop and contact the shift supervisor.

Never run a centrifugal pump against a closed suction valve. Never run the pump dry, even briefly, as the mechanical seal faces will be destroyed within seconds.

Stand clear of the coupling guard during starting. Hearing protection is required in the pump bay when any machine in the bay is running.

5. Tools and Materials

Routine starting requires no special tools. Have available a clean lint-free rag, the approved lubricant grade for top-up, a torch for sight glass inspection, hearing protection, and the shift log or work tablet for recording readings.

For a first start following maintenance, additionally obtain the handover documentation and the signed-off isolation certificate, and confirm with the shift supervisor that the permit is closed and every lock and tag has been removed from the driver and the switchgear.

6. Procedure - Startup

6.1 Walk down the pump and confirm the general condition: no loose fasteners, no oil on the baseplate, coupling guard secured, and no visible damage to instrument tubing or cabling.

6.2 Where the coupling was disturbed during maintenance, bump the motor uncoupled and verify rotation against the direction arrow before engaging the coupling and continuing with the start checks.

6.3 Check the bearing housing oil level at the sight glass. The level shall sit at the centre of the sight glass with the pump stopped. Top up with the approved lubricant grade if required.

6.4 Verify the suction strainer differential pressure gauge reads below the limit given in SOP-114. A blinded strainer at start is a common cause of immediate cavitation.

6.5 Open the suction valve fully. Crack the casing vent until a steady liquid stream is observed, confirming the casing is flooded and vapour free, then close the vent.

6.6 Confirm the seal flush line block valves are open and the seal pot level, where fitted, is between the marked limits. Confirm cooling or quench services to the bearing housings are lined up and flowing where fitted; a pump started without its quench can run correctly for minutes and then fail hours later, which is far harder to diagnose.

6.7 Close the discharge valve to the minimum-flow position marked on the valve stem, or fully closed where the pump has an automatic minimum-flow recirculation line.

6.8 Start the driver. Observe the discharge pressure gauge; pressure should rise promptly to the expected shut-in value. If pressure does not develop within ten seconds, stop the pump and investigate loss of prime.

6.9 Open the discharge valve slowly over thirty to sixty seconds while watching motor current, until the normal operating flow is established. Avoid running for more than two minutes at the closed-valve condition.

6.10 Check the mechanical seal area for leakage, feel the bearing housings for unusual warmth, and listen for cavitation, which presents as a sound like gravel passing through the pump.

6.11 Record the startup in the shift log, noting discharge pressure, motor current and any abnormal observation, and update the running-hours board for the pump train.

7. Procedure - Shutdown

7.1 For a planned shutdown, first close the discharge valve to roughly one-quarter open to reduce load, then stop the driver at the local station.

7.2 After the pump coasts down, close the discharge valve fully. Leave the suction valve open unless the pump is being handed over to maintenance.

7.3 If the pump is to be handed to maintenance, isolate and drain in accordance with SOP-118 and hang the appropriate tags before any work begins.

7.4 For standby pumps in autostart service, confirm the autostart selector is returned to the correct position and the pump remains flooded via the open suction valve.

7.5 Record the shutdown and the final running hours in the shift log, and note whether duty has transferred to the standby unit; if so, complete the standby start checks in full rather than assuming the machine will start on demand.

8. Acceptance Criteria

The pump is considered successfully started when discharge pressure and motor current are steady at their normal values, seal leakage is nil, and bearing housing temperature stabilises below 70 degrees C within thirty minutes. Any vibration judged abnormal shall be reported and assessed against SOP-110 before the pump is left in unattended service.

9. Records

Startup and shutdown events shall be recorded in the shift log. Abnormal observations shall be raised as work requests in the maintenance system quoting the pump tag and this procedure number.